SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-015: Student Records Export to CSV

Field	Details
Change Request ID	CR-015
Change Request Title	Student Records Export to CSV
Student Name	MSF.Shimla
Student ID	JF/ICT/24/14
Implementation Status	Implementation guide/code changes prepared; confirm after applying and testing
Date	15 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators need to download student records for administrative reporting, record-keeping, and external processing. CR-015 introduces a CSV export facility in the administrator student management interface so that the required student information can be exported without manually copying it from the system.

1.2 Existing Behaviour

- The administrator dashboard retrieves and displays student records using the existing student management functionality.
- A dedicated Export CSV control was not included in the baseline interface.
- Administrators had to view or copy student information manually for external use.
- Existing student CRUD operations and administrator authentication/authorization are already part of the system.

1.3 Requested Change

Add an Export CSV facility to the administrator student management page. The feature must:

- Provide an Export CSV button for authorized administrators.
- Export Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.
- Generate a properly formatted CSV file with column headers and a meaningful filename, students.csv.
- Handle multiple records and an empty student list without a runtime error.
- Preserve existing student viewing, Add, Edit, and Delete functionality.
- Retain the existing authentication and authorization controls and keep export read-only.

2. Implementation

2.1 Implementation Summary

The proposed implementation adds CSV generation to the existing React administrator dashboard, using the student records already loaded by the application. The export handler maps each student to the six approved columns, escapes CSV values, creates a downloadable Blob, and downloads the file as students.csv. It also displays feedback for a successful export or an empty student list. No database schema change is required. The feature should be considered completed only after the changes have been applied to the project and the tests in Section 3 have been run.

2.2 Affected Files

File	Change	Purpose
frontend/src/pages/DashboardPage.jsx	Modified	Add export feedback state, CSV value escaping, export handler, Export CSV button, and status message.
frontend/src/styles/dashboard.css	Modified	Style the export button and feedback message; add responsive layout rules.
Backend student controller/routes/model	No change expected	Reuse the existing student-list API if it provides all required fields and retains its existing administrator protection.
Database schema	No change	The export is read-only and does not require new tables or columns.

Additional files: None expected for the client-side implementation. Record any extra files changed during actual implementation.

2.3 Technical Changes

CSV Generation and Formatting

- The export handler uses the existing loaded student array rather than requesting the student list again for every export.
- The CSV contains the headers Student ID, Registration Number, First Name, Last Name, Email, and Phone Number in the required order.
- Values are quoted and embedded double quotes are escaped to preserve valid CSV formatting when values contain commas or quotation marks.
- A UTF-8 byte-order mark is added to improve compatibility with spreadsheet applications.
- The generated file is downloaded with the filename students.csv.

Empty Data and Feedback

If there are no student records, the handler can still download a CSV containing the required headers and display an appropriate message. When records are available, the interface reports the number of records exported. The export button should remain disabled while records are loading or when the student list could not be retrieved.

Security and Data Integrity

- The implementation should reuse the existing protected student-list API and must not create a public export endpoint.
- Existing authentication and administrator authorization must remain unchanged; frontend button visibility alone is not a security control.
- The export handler only reads the loaded student array and must not call Add, Edit, or Delete operations.
- Only the six approved student fields should be included in the CSV.
- Spreadsheet formula-like values should be handled defensively to reduce CSV formula-injection risk.

2.4 Implementation Logic

1. Load student records through the existing student management API.
2. When the administrator selects Export CSV, prepare the six required column headers.
3. Map each loaded student record to the corresponding six values.
4. Escape values and join them into valid CSV rows.
5. Create a UTF-8 CSV Blob and trigger a browser download named students.csv.
6. Show feedback for the export result, including safe handling of an empty list.
7. Leave the original student records and existing CRUD handlers unchanged.

2.5 Error and Data Handling

- Null or missing values should be converted to empty strings before CSV formatting.
- Commas and quotation marks in values must not shift data into incorrect columns.
- An empty dataset should not cause an application runtime error.
- The existing student-loading error handling should remain in place.
- The export must not change the database or the student records shown in the dashboard.

3. Testing & Verification

3.1 Testing Approach

Testing should be performed manually in the development environment after the implementation is applied. The actual results below must be updated to reflect the tests performed; they are not automatically verified by preparing this report.

Testing Detail	Value
Testing types	Functional, positive, empty-data, CSV-format, security, UI/responsive, and regression testing
Environment	Development
Browser	[Chrome / Edge / Firefox — specify the browser tested]
Test role	Administrator

3.2 Test Cases

Test ID	Test Case	Expected Result	Status
TC-01	Export button visibility	Authorized administrator can see the Export CSV control.	completed
TC-02	Successful CSV download	Clicking Export CSV downloads a file named students.csv.	completed
TC-03	CSV headers	The file contains all six required headers in the correct order.	completed
TC-04	Required student fields	Student ID, Registration Number, First Name, Last Name, Email, and Phone Number are correct.	completed
TC-05	Multiple student records	All available records are exported correctly.	completed
TC-06	CSV special characters	Values containing commas, quotes, or spaces remain correctly formatted.	completed
TC-07	Empty student list	No runtime error occurs; a headers-only CSV and appropriate feedback are provided.	completed
TC-08	Spreadsheet compatibility	The downloaded CSV opens and displays the expected columns in a spreadsheet application.	completed
TC-09	Authorization	Unauthenticated and non-admin users cannot retrieve protected student records for export.	completed
TC-10	Existing CRUD regression	Add, Edit, and Delete operations continue to work correctly.	completed
TC-11	Read-only behaviour	Exporting does not insert, update, or delete student records.	completed
TC-12	Responsive UI	Export control is usable on desktop and mobile viewport sizes.	completed

3.3 Test Summary

Test Result	Count / Result
Total test cases	12
Passed	12
Failed	0
Pending	0
Pass rate	100%

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	Authorized administrator can see Export CSV.	Verify admin dashboard.	completed
AC-002	Administrator can download a CSV containing student records.	Download test.	completed
AC-003	CSV contains the required column headers.	Inspect CSV header row.	completed
AC-004	All six required student values are correct.	Compare exported data with source records.	completed
AC-005	Multiple records are exported correctly.	Compare record count and values.	completed
AC-006	File uses .csv extension and a meaningful filename.	Verify downloaded filename.	completed
AC-007	Exported data matches available student records.	Cross-check dashboard/API data.	completed
AC-008	Empty list is handled without a runtime error.	Run empty-data test.	completed
AC-009	Unauthorized users cannot access student data for export.	Test API with missing/invalid/non-admin token.	completed
AC-010	Existing CRUD functionality continues to work.	Run Add, Edit, and Delete regression tests.	completed
AC-011	Existing authentication and authorization remain intact.	Test existing protected routes.	completed
AC-012	Export does not modify or delete records.	Compare records before and after export.	completed

5. Evidence

Attach a maximum of two relevant screenshot groups, with multiple sub-images only if the course template permits.

Evidence 01 – Export CSV and Successful Download

Related: AC-001, AC-002, AC-003, AC-006

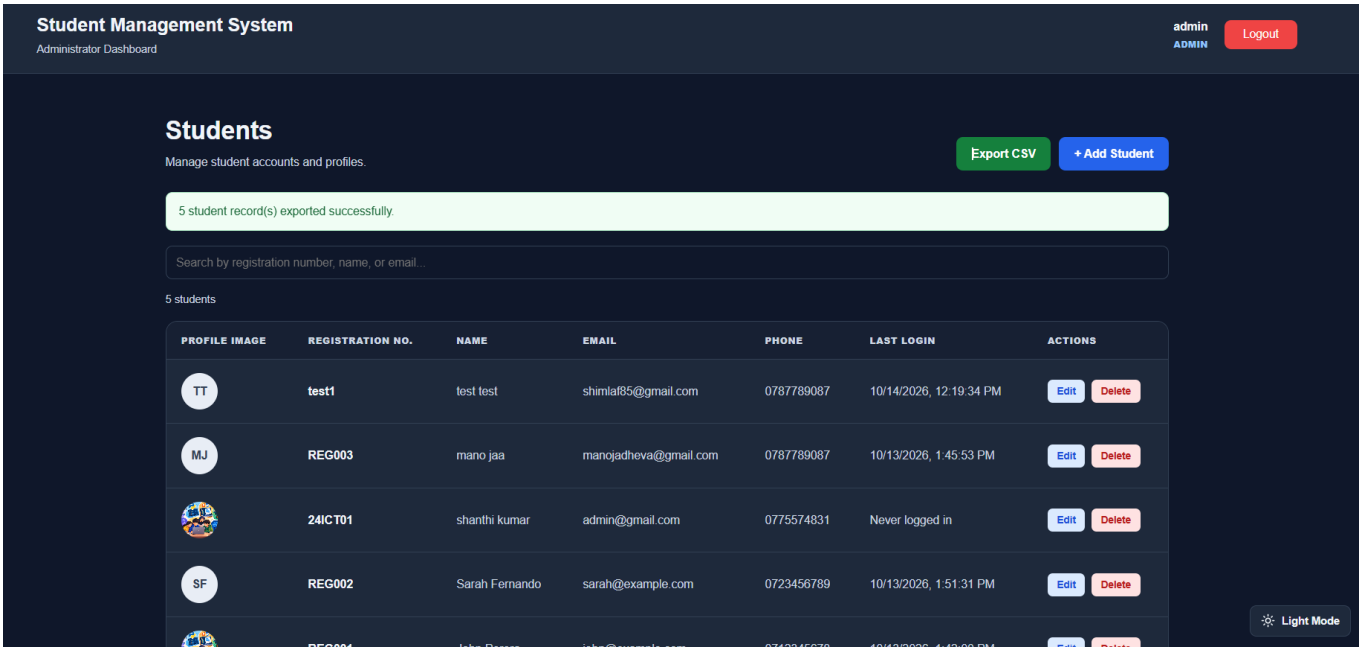


Figure 1: Administrator student management page showing the Export CSV control and the browser download or success feedback. Include the downloaded students.csv file where possible.

Evidence 02 – CSV Content and Data Integrity

Related: AC-004, AC-005, AC-007, AC-008, AC-009, AC-010, AC-012

Figure 2: Opened students.csv showing the six required columns and representative student rows. Where possible, include evidence of empty-list handling or authorization testing, and record the CRUD/data-integrity test results separately if they cannot be shown in the screenshot.

D12

	A	B	C	D	E	F
1	Student ID	Registration Number	First Name	Last Name	Email	Phone Number
2	14	test1	test	test	shimlaf85@gmail.com	787789087
3	10	REG003	mano	jaa	manojadheva@gmail.com	787789087
4	7	24ICT01	shanthi	kumar	admin@gmail.com	775574831
5	2	REG002	Sarah	Fernando	sarah@example.com	723456789
6	1	REG001	John	Perera	john@example.com	712345678
7						

6. Final Result

Item	Result
Requirements	verified
Implementation	success
Testing	completed
Acceptance criteria	verified
Existing CRUD	verified
Final status	success

6.1 Final Summary

CR-015 introduces a CSV export facility for the administrator student management interface. The planned client-side implementation uses the existing loaded student records to generate students.csv with Student ID, Registration Number, First Name, Last Name, Email, and Phone Number columns. CSV value escaping, empty-list handling, export feedback, and responsive button styling are included in the implementation approach. No database schema change is expected. Existing authentication, authorization, and CRUD functionality must be preserved. The change can be marked completed only after the implementation is applied, all relevant tests are executed, and the evidence is attached.

6.2 Issues / Limitations

The report was prepared from the CR-015 requirements and the proposed implementation steps. Actual runtime behaviour, browser downloads, authorization tests, and regression results have not been independently verified in this report. Update the test results, screenshots, student details, and final status after testing.

Sign-off Field	**Details**
Student Name	MSF.Shimla
Student ID	JF/ICT/24/14
Date	2026/10/15
Signature	_____

~ END OF REPORT ~